

Travelling NAMU WIKI

:Navigating Open-Domain Knowledge Graphs via Reinforcement Learning

1. Introduction to the Task

We introduce the **"Wiki-Speedrun" task**, where an agent must navigate from a random source document to a target document **in the fewest possible steps**. Given the massive scale of the document corpus and the hidden underlying graph structure, the agent must rely entirely on its understanding of **semantic topology** and sequentially exploit cues from available hyperlinks. We anticipate that this task will serve as an effective proxy for reasoning in semantically rich contexts, such as Multi-hop RAG.

2 Problem Formulation (MDP)

To address computational overhead while preserving local exploration capability, we formulate the task as a Markov Decision Process (MDP) relying strictly on the current semantic context:

- **State Space (S):** The state s_t is defined by concatenating the text embeddings of the current document and the target document's title: $s_t = [e_{curr\ doc}; e_{target\ title}]$.
- **Action Space (A):** The action a_t is the selection of a specific hyperlink present in the current document. Each candidate is represented by the embedding of its anchor text. Crucially, the size of this action space is highly variable across different documents.
- **Reward (R):** A sparse signal designed to encourage the shortest path: e.g., +100 for reaching the target, -10 for failing (dead-end or time-out), and a step penalty of -1.

3. Project Design

3.1. Objective of this Project

A key challenge of this task is that, unlike pathfinding in a small gridworld, the agent must navigate through a massive number of documents to reach the target, inherently resulting in extremely sparse rewards.

To address these challenges, we will explore the optimal action and reward formulations for this task, and ultimately investigate effective reinforcement learning methodologies to solve this problem.

Specifically, we aim to demonstrate that (1) while a simple greedy semantic search may perform well to a certain extent, it ultimately fails to reach the target when paths become excessively long or require detour-heavy trajectories, and (2) applying advanced learning algorithms designed to overcome sparse rewards, such as HER (Andrychowicz et al., 2017), can effectively resolve this limitation.

3.2. High-level Architecture

To naturally handle the variable action space, we propose a Cross-Attention Actor-Critic Architecture. Rather than outputting a fixed number of logits, the state vector s_t acts as the Query (Q), while the N candidate anchor embeddings in the current document act as the Keys (K) and Values (V). The policy probability $\pi(a_i | s_t)$ is computed via a masked softmax over the attention scores.

This allows the policy network to dynamically evaluate an arbitrary number of semantic links simultaneously. The context vector from the attention mechanism is then passed to a Critic network to estimate the state value $v(s_t)$.

3.3. Experimental Setup

- **Baselines**
 - **Random Walk:** Randomly selects an available hyperlink at each step.
 - **Heuristic Search:** Greedy selection or Beam Search based on local semantic similarity to the target.
 - **Standard PPO:** Baseline PPO trained without HER.
 - **PPO + HER:** Our proposed approach utilizing Hindsight Experience Replay.
- **Evaluation Metrics**
 - **Success Rate (SR):** The percentage of episodes that successfully reach the target document within a given step limit.
 - **Average Path Length:** The mean number of steps taken in successful trajectories.

4. Related Works

Each cluster of prior work informs a specific design choice or motivates a specific challenge in our Wiki-Speedrun setup.

RL for Graph and Web Navigation. Our task is most directly related to RL agents that learn to navigate hyperlinked corpora. Nogueira and Cho (2016) first cast goal-driven web navigation as an RL problem, and Gur et al. (2019) extended this to richer web pages with structured DOMs—together they establish that link-selection policies can be trained end-to-end, which justifies our action interface. Zaheer et al. (2022) is the closest precedent for Wikipedia-scale hyperlink navigation, showing that behavioral cloning from randomly sampled walks can train an effective link-selection policy. This validates our state design $[e_curr_doc ; e_target_title]$. M-Walk (Shen et al., 2018) and Pisacane et al. (2024) target graph path-finding under partial observability and sparse signals, motivating our hypothesis that purely greedy semantic policies will fail on detour-heavy trajectories and that learned value estimation is required. Yan et al. (2025) demonstrates multi-turn RL for screen-level navigation and provides a recent recipe for stabilizing long-horizon exploration that our PPO setup builds on.

Variable / Large Action Spaces. Because the number of hyperlinks differs at every node, we cannot use a fixed-size policy head. He et al. (2016) introduced Deep Reinforcement Relevance Networks for natural-language action spaces, which directly motivates our cross-attention actor that scores each anchor embedding against the state. Dulac-Arnold et al. (2015) addresses scalability when the per-step action set is enormous, supporting our choice to operate in embedding space rather than over a one-hot vocabulary of all wiki pages.

Goal-Conditioned RL under Sparse Reward. The $+100 / -1$ reward structure makes vanilla policy gradients data-inefficient. Hindsight Experience Replay (Andrychowicz et al., 2017) re-labels failed trajectories with achieved goals so that every rollout produces useful supervision. For Wiki-Speedrun, every dead-end episode can be relabeled as a successful trajectory toward the document the agent actually landed on—this is the central mechanism we hypothesize will close the gap between greedy heuristics and learned policies.

Yan, Haolong, et al. "GUI Exploration Lab: Enhancing Screen Navigation in Agents via Multi-Turn Reinforcement Learning." Advances in Neural Information Processing Systems 38 (2025).

Gur, Izzeddin, et al. "Learning to Navigate the Web." International Conference on Learning Representations (2019).

Nogueira, Rodrigo, and Kyunghyun Cho. "End-to-End Goal-Driven Web Navigation." *Advances in Neural Information Processing Systems* 29 (2016).

Zaheer, Manzil, et al. "Learning to navigate wikipedia by taking random walks." *Advances in Neural Information Processing Systems* 35 (2022).

Shen, Yelong, et al. "M-walk: Learning to walk over graphs using monte carlo tree search." *Advances in Neural Information Processing Systems* 31 (2018).

Pisacane, Alexei, Victor-Alexandru Darvari, and Mirco Musolesi. "Reinforcement Learning Discovers Efficient Decentralized Graph Path Search Strategies." *arXiv preprint arXiv:2409.07932* (2024).

He, Ji, et al. "Deep reinforcement learning with a natural language action space." *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 2016.

Dulac-Arnold, Gabriel, et al. "Deep reinforcement learning in large discrete action spaces." *arXiv preprint arXiv:1512.07679*(2015).

Andrychowicz, Marcin, et al. "Hindsight experience replay." *Advances in neural information processing systems* 30 (2017).